

MAT 425 Counting Report

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1 Introduction

This report includes some proofs useful for the construction of spherical harmonics in p dimensions. We prove two Lemmas to be used in Proposition 4.2 and Theorem 4.4 as taken from the article written by Christopher Frye and Costas Efthimiou [1]. Proposition 4.2 and Theorem 4.4 both enumerate the quantity of linearly independent homogeneous polynomials given the degree and number of variables, and Theorem 4.4 relates to polynomials that are also harmonic.

2 Lemma 4.1

Lemma 4.1 states that for $0 < |r| < 1$,

$$\frac{1}{(1-r)^p} = \sum_{j=0}^{\infty} \frac{(p+j-1)!}{j!(p-1)!} r^j.$$

Proof. We will use a counting trick to prove this. First recall that when $0 < |r| < 1$, the sum of a geometric series is given by:

$$\frac{1}{1-r} = \sum_{n=0}^{\infty} r^n.$$

Since $0 < |r| < 1$, we can write $(1-r)^{-p}$ in terms of a product of geometric series, i.e.,

$$\begin{aligned} (1-r)^{-p} &= \left(\frac{1}{1-r}\right) \left(\frac{1}{1-r}\right) \dots \left(\frac{1}{1-r}\right) (p \text{ times}) = \\ &= \left(\sum_{n=0}^{\infty} r^n\right) \left(\sum_{n=0}^{\infty} r^n\right) \dots \left(\sum_{n=0}^{\infty} r^n\right) (p \text{ times}). \end{aligned}$$

From this we can compute the *Cauchy Product* of the p infinite series. The result will be an infinite series with a constant term and all positive integer

powers of r ,

$$(1 - r)^{-p} = \sum_{j=0}^{\infty} c_j r^j.$$

In order to compute the coefficients c_j 's we must count how many different ways we can produce r^j in the multiplication. To do this we will use a counting trick referred to as *Stars and Bars* in combinatorics. The idea behind this trick is asking how many ways can we put j indistinguishable balls into p distinguishable boxes? This is given by “ $p - j + 1$ choose j ”,

$$\binom{p - 1 + j}{j} = \frac{(p - 1 + j)!}{j!(p - 1 + j - j)!} = \frac{(p - 1 + j)!}{j!(p - 1)!},$$

and thus the lemma is proved. $\square$

3 Proposition 4.2

Proposition 4.2 states that

$$K(p, n) = \frac{(n + p - 1)!}{n!(p - 1)!}$$

This indicates how many linearly independent homogeneous polynomials of degree n in p variables there are.

Proof. We are trying to determine how many distinct monomials of degree n can be created with p variables. A homogeneous polynomial of degree n is a sum of terms in which the exponents of the variables of each term sum to n . For instance, a homogeneous polynomial of degree 5 in the variables x , y , and z can have terms like xy^4 , or x^3z^2 , or y^2z^3 but not x^2y or xz^2 because the degree does not equal n .

In order to determine the number of these monomials, we need to count the number of ways we can assign integer exponents (that are not negative) to p variables so that their sum is n . This can be shown by the following equation:

$$a_1 + a_2 + \cdots + a_p = n$$

Here, $a_1, a_2, \dots, a_p$ represent the exponents of our variables. To determine this type of polynomial, we must give the coefficient of every possible n -th degree monomial. To find $K(p, n)$, we must count how many possible n -th degree monomials there are in p variables. This is a question that can be easily imagined as “How many ways can we place n indistinguishable rocks into p distinguishable bins?”

The number of ways to do this is given by:

$$\binom{n + p - 1}{p - 1} = \binom{n + p - 1}{n} = \frac{(n + p - 1)!}{n!(p - 1)!}$$

This counts how many non-negative integer solutions to the equation above exist. Thus, the number of distinct monomials of degree n in p variables is $\binom{n+p-1}{n}$, proving the theorem. $\square$

4 Lemma 4.3

Lemma 4.3 states that

$$\binom{k}{l} = \frac{k}{l} \binom{k-1}{l-1}$$

In order to prove this expand the right side of the equation and compute:

$$\begin{aligned} \frac{k}{l} \binom{k-1}{l-1} &= \frac{k}{l} \cdot \frac{(k-1)!}{(l-1)!(k-l)!} \\ &= \frac{k \cdot (k-1)!}{l \cdot (l-1)!(k-l)!} = \frac{k!}{l!(k-l)!} = \binom{k}{l} \\ \therefore \binom{k}{l} &= \frac{k}{l} \binom{k-1}{l-1} \end{aligned}$$

5 Theorem 4.4

Theorem 4.4 states that if $N(p, n)$ is the number of linearly independent homogeneous polynomials of degree n with p variable, then

$$N(p, n) = \frac{2n + p - 2}{n} \binom{n + p - 3}{n - 1}$$

This means that given the degree and the number of variables, $N(p, n)$ linearly independent variables would form a basis for the space.

Proof. To begin the proof for this theorem, let H_n be a homogeneous harmonic polynomial of degree n in p variables.

$$H_n(x_1, x_2, \dots, x_p) =$$

$$(a_{11}x_p^n) + (a_{21}x_p^{n-1}x_1 + \dots + a_{2p}x_p^{n-1}x_{p-1}) + \dots + (a_{n1}x_1^n + \dots + a_{np}x_{p-1}^n)$$

Where $a_{ij} \in \mathbb{R}$ Notice that each grouping of parentheses share the same exponent of x_p , which may be factored out

$$= x_p^n(a_{11}) + x_p^{n-1}(a_{21}x_1 + \dots + a_{2p}x_{p-1}) + \dots + x_p^0(a_{n1}x_1^n + \dots + a_{np}x_{p-1}^n)$$

Consider that the j th term is x_1^j times some homogeneous polynomial of degree $n-j$ with variable from x_1 to x_{p-1} . Call this polynomial $h_{n-j}(x_1, x_2, \dots, x_{p-1})$

$$\therefore H_n(x_1, x_2, \dots, x_p) = \sum_{j=0}^n x_p^j h_{n-j}(x_1, x_2, \dots, x_{p-1})$$

Since H_n is harmonic, then:

$$\begin{aligned} 0 &= \Delta H_n = \left(\frac{\delta^2}{\delta x_p^2} + \Delta_{p-1} \right) H_n = \frac{\delta^2}{\delta x_p^2} H_n + \Delta_{p-1} H_n \\ &= \frac{\delta^2}{\delta x_p^2} \sum_{j=0}^n x_p^j h_{n-j}(x_1, x_2, \dots, x_{p-1}) + \Delta_{p-1} \sum_{j=0}^n x_p^j h_{n-j}(x_1, x_2, \dots, x_{p-1}) \end{aligned}$$

Since the Laplace operator is commutative over addition

$$= \sum_{j=0}^n \frac{\delta^2}{\delta x_p^2} (x_p^j h_{n-j}(x_1, x_2, \dots, x_{p-1})) + \sum_{j=0}^n \Delta_{p-1} (x_p^j h_{n-j}(x_1, x_2, \dots, x_{p-1}))$$

Since h_{n-j} contains no x_p terms it is treated as constant in the double partial derivative and $\frac{\delta^2}{\delta x_p^2} (x_p^j) = \frac{\delta}{\delta x_p} (j)(x_p^{j-1}) = (j)(j-1)x_p^{j-2}$. Define $h_{-1} = h_{-2} = 0$, which removes the terms with x_p^0 and $x_p^1/$

$$= \sum_{j=2}^n (j)(j-1)x_p^{j-2} h_{n-j} + \sum_{j=0}^n \Delta_{p-1} (x_p^j h_{n-j})$$

Reindex the first sum and combine the sums results in:

$$\begin{aligned} &= \sum_{j=0}^n (j+2)(j+1)x_p^{j-2} h_{n-j-2} + \Delta_{p-1} (x_p^j h_{n-j}) \\ &= \sum_{j=0}^n x_p^j ((j+2)(j+1)h_{n-j-2} + \Delta_{p-1} h_{n-j}) \end{aligned}$$

Since each x_p^j is linearly independent due to having different degrees, it must be true that the coefficients are all equal to 0. For each j and x_p^j there is a coefficient of the form $j(j-1)h_{n-j-2} + \Delta_{p-1} h_{n-j}$, substituting in value of $j = 1$ to $j = n$ creates the system of equations:

$$\begin{aligned} j = 0 &\mapsto 2h_{n-2} + \Delta_{p-1} h_n = 0 \\ j = 1 &\mapsto 6h_{n-3} + \Delta_{p-1} h_{n-1} = 0 \\ &\dots \\ j = n-2 &\mapsto n(n-1)h_0 + \Delta_{p-1} h_2 = 0 \\ j = n-1 &\mapsto \Delta_{p-1} h_1 = 0 \\ j = n &\mapsto \Delta_{p-1} h_0 = 0 \end{aligned}$$

All h_{n-j} may be defined in terms of either h_n or h_{n-1} . Given h_n , the first equation allows h_{n-2} to be defined, which may be used to define h_{n-4} and the pattern continues for all h_{n-j} where j is even. Given h_{n-1} the same pattern

holds to define all h_{n-j} where j is odd. Thus the number of possible h_{n-j} s is determined by the number of possible h_n s and h_{n-1} s.

By Proposition 4.2 h_n can be written in terms of $K(p-1, n)$ basis polynomials since h_n only has $p-1$ variables and similarly h_{n-1} can be written in terms of $K(p-1, n-1)$ basis polynomials. Thus to determine h_n and h_{n-1} the coefficient for the linear combination of these basis polynomials must be decided. This amounts to $K(p-1, n) + K(p-1, n-1)$ total coefficients required to determine any h_n, h_{n-1} , which is all that is required to determine any H_n .

Therefore,

$$N(p, n) = K(p-1, n) + K(p-1, n-1) = \binom{p+n-2}{2} + \binom{p+n-3}{n-1}$$

Using Lemma 4.3, this can be written as such

$$\begin{aligned} N(p, n) &= \frac{p+n-2}{n} \binom{p+n-2-1}{n-1} + \binom{p+n-3}{n-1} \\ &= \left(\frac{p+n-2}{n} + 1 \right) \binom{p+n-3}{n-1} = \frac{2n+p-2}{n} \binom{n+p-3}{n-1} \end{aligned}$$

Thus concluding the proof. $\square$

6 References

References

- [1] Christopher Frye and Costas Efthimiou. "Spherical Harmonics In p Dimensions". In: *Department of Physics* (2012), p. 95.